Assignment 3(2012)(24-25 Second)

Fondamental concepts of math (The University of Hong Kong)

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A3/MATH2012/2024-25/2nd

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2012 Fundamental Concepts of Mathematics

Assignment 3

Due date : Mar 26, 2025 before 12:00 noon.

Instructions

- Please scan and submit your work as a **PDF** file in **moodle**. No late work will be accepted.

  Warning: Make sure that your work is properly scanned. **Oversized, blurred or upside-down images will NOT be graded**.

Guidelines on Writing

- You should always give precise and adequate explanations to support your conclusions. Note that being able to write clearly is as important a mathematical skill as being able to solve equations. Putting an idea on paper requires careful thought and attention. Writing forces you to express your ideas clearly and in the proper order.
- Proper writing of mathematics may not have been emphasized in your previous mathematics courses, but as you start learning university-level mathematics, you may have to adjust your approach to doing mathematics.
- Suggestions:
  - Write in complete sentences. Do not just give the solution in fragmentary bits and pieces. Clarity of presentation of your argument counts.
  - Explain the meaning of every symbol that you introduce and avoid starting a sentence with a symbol.
  - Use the implication symbol "$\Longrightarrow$" and the "if and only if" symbol "$\Longleftrightarrow$" correctly.
  - Be careful to use quantifiers "for all", "for any", "for some", "there exists" correctly. You cannot expect the reader to understand which you mean if you omit these quantifiers.
  - Avoid using the symbols "$\because$" and "$\therefore$". Use "because/since", "therefore/hence/it follows that".

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References on Mathematical Writing

- Chapter 0 of *Mathematical Proofs: A Transition to Advanced Mathematics*
- Section I (Chapter 2 - 4) of *How to Think Like a Mathematician: A Companion to Undergraduate Mathematics*
- Y.M. Chan, *Writing Mathematics*

Remark

- You are welcome to contact the instructor or the tutor(s) if you have any difficulties. See '*Course Information*' for availabilities.

You need only to give your answers to those questions indicated with *. Other problems are optional and will not be marked. However, we do expect you to eventually finish all problems.

Questions with * in this assignment:

Q.3, Q.6, Q.8, Q.10, Q.11(i), (k), (q), Q.12, Q.15, Q.22, Q.24, Q.27.

Chapter 3 (Section 3.4)

1. Prove that there does not exist an integer x such that $4x \equiv 2 \pmod{28}$.

2. Prove that for any $a \in \mathbb{R}$ with $a \geq 0$, if $a < \mu$ for any $\mu > 0$, then $a = 0$.

*3. Prove that for any $x, y \in \mathbb{R}$,

   (a) if $x \in \mathbb{Q}$ and $y \in \mathbb{I}$ (the set of irrational numbers), then $x + y \in \mathbb{I}$;

   (b) if $x \in \mathbb{Q} \setminus \{0\}$ and $y \in \mathbb{I}$, then $xy \in \mathbb{I}$.

4. Prove that there does not exist two nonzero perfect squares m and n such that their difference is 1 (an integer m is a *perfect square* if $m = k^2$ for some $k \in \mathbb{Z}$).

5. Prove that for any subsets A and B of a universal set U,

$$B = A^c \quad \text{if and only if} \quad A \cup B = U \text{ and } A \cap B = \emptyset.$$

*6. Prove that for any $m, n \in \mathbb{Z}$, if $mn \not\equiv 0 \pmod 3$, then $m^2 + n^2$ cannot be a perfect square.

7. Prove that there do not exist integers $m \geq 2$ and $n \geq 1$ such that $m^2 + 1 = 2^n$.

*8. (a) Prove that for any $n \in \mathbb{Z}$, $n \equiv 0 \pmod 3$ if and only if $n^2 \equiv 0 \pmod 3$.

   (b) Prove that $\sqrt{3}$ is irrational. (*Hint*: The proof is similar to that of *Example 3.25* in the lecture note. You may also need to use the result from (a).)

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9. Prove that for any $n \in \mathbb{N}$ with $n \geq 2$, $\sqrt{n}$ is irrational if and only if n is not a perfect square.

   (*Hint*: You may need to use the *Fundamental Theorem of Arithmetic*: every integer $n \geq 2$ can be written as a product of one or more (not necessarily distinct) primes, that is, $n = p_1 p_2 \cdots p_r$ for some prime numbers $p_1, p_2, \ldots, p_r$.)

*10. Prove that there do not exist positive integers m, n such that $\sqrt{m} + \sqrt{n} = 2012$ and m and n are not perfect squares.

   (*Hint*: You can use the result from **Q.9**.)

Chapter 3 (Section 3.5)

11. Prove or disprove each of the following statements:

   (a) $\forall a, b \in \mathbb{Q}$, a^b is rational.

   (b) There exists distinct positive integers such that $\sqrt{a+b} = \sqrt{a} + \sqrt{b} - 2$.

   (c) For any $a, b, c, d \in \mathbb{R}$, if $a < b$ and $c < d$, then $ac < bd$.

   (d) For any nonempty sets A, B and C, $A \setminus (B \setminus C) = (A \setminus B) \setminus C$.

   (e) For any nonempty sets A, B and C, if $A \times B \subseteq A \times C$, then $B \subseteq C$.

   (f) For any sets A and B, $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$.

   (g) There exists $n \in \mathbb{Z}$ such that $n^2 + 2 \equiv 0 \pmod 4$.

   (h) There exist $a, b \in \mathbb{Z}$ and there exists $n \in \mathbb{N}$ with $n \geq 2$ such that $a \not\equiv b \pmod n$ and $a^2 \equiv b^2 \pmod n$.

   *(i) For any $n \in \mathbb{N}$, if $2^n - 1$ is prime, then n is prime.

   (j) Every odd integer is the sum of three odd integers.

   *(k) $\exists N \in \mathbb{N}$, $\forall n \in \mathbb{N}$ with $n \geq N$, $n^2 - n + 11$ is prime.

   (l) $\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$, $e^y - x > 0$.

   (m) $\exists y \in \mathbb{R}$, $\forall x \in \mathbb{R}$, $e^y - x > 0$.

   (n) For any $m, n \in \mathbb{Q}$ with $m < n$, there exists $r \in \mathbb{Q}$ such that $r \in (m, n)$.

   (o) There exist three distinct positive integers m, n, p such that $m^n = n^p$.

   (p) For any $n \in \mathbb{N}$, if $n = ab$ for some $a, b \in \mathbb{N}$, then $a \geq \sqrt{n}$ or $b \geq \sqrt{n}$.

   *(q) There exist two prime numbers whose difference is 151.

   (r) For any positive odd integer $m \geq 3$, there exists $n \in \mathbb{N}$ such that $n^2 + nm$ is a perfect square.

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*12. Prove or disprove: For any prime number $p \geq 5$, $p^2 \equiv 1 \pmod{24}$.

The following solution is generated by **ChatGPT** (with slight modification of wordings). If you think you should prove the given statement, comment and find out the missing gap(s) in the following proof (if any). If you think you should disprove the given statement, point out the mistake(s) in the following solution and provide a disproof.

"Proof": Any prime number greater than 3 can be written in the form $6k \pm 1$ for some nonnegative integer k. This is because any integer n can be written in one of the forms $6k$, $6k+1$, $6k+2$, $6k+3$, $6k+4$, $6k+5$ (or $6k-1$) for some integer k, and the primes among them must be of the form $6k \pm 1$.

Now, consider the square of a prime number of the form $6k \pm 1$. We have

$$(6k \pm 1)^2 \;=\; 36k^2 \pm 12k + 1 \;=\; 12(3k^2 \pm k) + 1.$$

Since $3k^2 \pm k$ is always an integer, we must have $(6k \pm 1)^2 \equiv 1 \pmod{24}$. Therefore, for any prime number $p \geq 5$, we have $p^2 \equiv 1 \pmod{24}$.

Chapter 3 (Section 3.6 – Mathematical Induction)

13. **(Sum of a Geometric Sequence)**

Prove that for any integer $n \geq 0$ and any real number $r \neq 1$, we have

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}.$$

14. **(Generalization of De Morgan's Law)**

Prove that for any subsets $A_1, A_2, \ldots, A_n$ of a universal set U and for any integer $n \geq 2$,

$$(A_1 \cap A_2 \cap \cdots \cap A_n)^c \;=\; A_1^c \cup A_2^c \cup \cdots \cup A_n^c.$$

*15. Prove that $17^n - 10^n$ is divisible by 7 for any $n \in \mathbb{N}$.

16. Prove that for any $n \in \mathbb{N}$, we have

$$4^n > n^3.$$

(*Hint*: Show that the inequality is true for $n = 1$ and use induction for $n \geq 2$.)

17. Prove that for any $q \in \mathbb{R}$ with $0 < q < 1/2$,

$$(1+q)^n \leq 1 + 2^n q$$

for any $n \in \mathbb{N}$.

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18. Prove that for any $n \in \mathbb{N}$,

$$\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^n} \geq 1 + \frac{n}{2}.$$

19. Consider the sequence $F_1, F_2, F_3, \ldots$ of *Fibonacci numbers* satisfying $F_1 = 1$, $F_2 = 1$ and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for any integer } n \geq 3.$$

Prove that

$$F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1} \quad \text{for any } n \in \mathbb{N}.$$

Chapter 3 (Section 3.6 – Strong Form of Induction)

20. A sequence $x_1, x_2, x_3, \ldots$ of numbers is defined by $x_1 = 1$, $x_2 = 4$, $x_3 = 9$ and

$$x_n = x_{n-1} - x_{n-2} + x_{n-3} + 2(2n-3)$$

for any integer $n \geq 4$. Guess a formula for x_n and prove that your guess is correct.

21. A sequence $x_1, x_2, x_3, \ldots$ of numbers is defined by $x_1 = 1$ and $x_2 = 1$ and for any integer $n \geq 3$, x_n is the sum of all of the previous terms in the sequence. Guess a formula for x_n and prove that your guess is correct.

*22. Let $F_1, F_2, F_3, \ldots$ be the sequence of Fibonacci numbers (defined in **Q.19** above). Prove that for any $n \in \mathbb{N}$,

$$F_n \text{ is even if and only if } n \equiv 0 \pmod 3.$$

23. Let $F_1, F_2, F_3, \ldots$ be the sequence of Fibonacci numbers (defined in **Q.19** above). Prove that for any integer $n \geq 2$,

$$F_n^2 - F_{n+1}F_{n-1} = (-1)^{n+1}.$$

(*Hint*: The statement can be proved by either the ordinary induction or the strong form.)

*24. Let $F_1, F_2, F_3, \ldots$ be the sequence of Fibonacci numbers (defined in **Q.19** above). Prove that for any $n \in \mathbb{N}$,

$$F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}, \quad \text{where } \alpha = \frac{1+\sqrt{5}}{2} \text{ and } \beta = \frac{1-\sqrt{5}}{2}.$$

(*Hint*: Note that α and β are the two roots of the quadratic equation $x^2 = x + 1$.)

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25. **[Past Exam Question (May 2022 Q.3)]**
Let $F_1, F_2, F_3, \ldots$ be the sequence of Fibonacci numbers (defined in **Q.19** above).

(a) Find all $n \in \mathbb{N}$ such that $F_n = n^2$.
(*Hint*: Use induction to show that $F_n > n^2$ from a certain value $n = n_0$ onward.)

(b) Let $\alpha = \frac{1+\sqrt{5}}{2}$. Express α^n in terms of α, F_{n-1} and F_n for any integer $n \geq 2$, and use induction to prove that your formula is correct.
(*Hint*: Note that α is a root of the quadratic equation $x^2 - x - 1 = 0$.)

(c) Use the results from (a) and (b) to find all $n \in \mathbb{N}$ such that $\alpha^n - n^2\alpha$ is an integer.
(*Hint*: You may use without proof the fact that α is irrational.)

26. Let $F_1, F_2, F_3, \ldots$ be the sequence of Fibonacci numbers (defined in **Q.19** above). prove that for any $m, n \in \mathbb{N}$,

$$F_{m+n+1} = F_{n+1}F_{m+1} + F_nF_m.$$

*27. **(The postage stamp problem)**

Determine which postage amounts can be created using the stamps of 3 and 7 cents. In other words, determine the exact set of positive integers n such that each of them can be written in the form $3x + 7y$ for some nonnegative integers x and y.

(*Hint*: Check the first few values of n directly. Then use the strong form of induction to show that, from a certain value n_0 onwards, all numbers n have such a representation.)

28. **(Fundamental Theorem of Arithmetic)**

Prove that every integer $n \geq 2$ can be written as a product of one or more (not necessarily distinct) primes, that is, $n = p_1p_2 \cdots p_r$ for some prime numbers $p_1, p_2, \ldots, p_r$.

(*Hint*: In the inductive step, if $k+1$ is prime, then we are done. Otherwise, you may need to use the following definition of composite numbers: An integer $n \geq 2$ is called *composite* if it can be factored as $n = ab$ for some integers a and b satisfying $2 \leq a, b < n$.)

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